

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Determine if the series is absolutely convergent, conditionally convergent, or divergent.

a) $\sum_{n=1}^{\infty} (-1)^n \frac{2}{\sqrt[3]{n}}$

b) $\sum_{n=2}^{\infty} (-1)^n n e^{-n}$

c)
$$\sum_{n=1}^{\infty} \frac{\sqrt{n \arctan(n)}}{n^{1/2}}$$

d)
$$\sum_{n=0}^{\infty} \frac{n^2 - n + 1}{\sqrt{n^5 + 300n^3 + 2}}$$

2. Is it true that if $\sum_{n=1}^{\infty} a_n$ is divergent then $\sum_{n=1}^{\infty} |a_n|$ is also divergent?

3. What if $\sum_{n=1}^{\infty} |a_n|$ is convergent; is $\sum_{n=1}^{\infty} a_n$ also convergent?

4. If $\sum_{n=1}^{\infty} a_n$ is convergent is $\sum_{n=1}^{\infty} |a_n|$ also convergent?

5. Consider the series $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2}$

a) Show that the given series is convergent.

b) While we know the given series is convergent we don't know its final value. Find s_{10} to use it as an estimate of what the series converges to.

- c) Clearly the series doesn't converge to the value of s_{10} , but it should converge to something close to this value. Find an upper bound on the error of the approximation of s_{10} as the sum of the series.
- d) Suppose that we wanted to estimate the sum of the series accurately to within $\frac{1}{1,000,000}$ of its actual value. How many terms of the series would we need to use in a partial sum of the series to do this?
- e) It can be shown that the exact sum of this series is $-\frac{\pi^2}{12}$. Use Wolfram Alpha to show that the differences between your answers to parts (c) and (d) and the actual sum of the series are indeed accurate to the appropriate level.